

Greek-Neutral Hedging Strategies

A Comparative Backtesting Study under Market Frictions

Group 10

Course: Quantitative Risk Management (DBA5109)

Instructor: Dr. Yifan Feng

Sai Ashwin Kumar Chandramouli (A0329294U)

Hrithik Kannan Krishnan (A0318899B)

Zihan Zhou (A0318419Y)

Oishika Saha (A0318298N)



Motivation & Research Question

Why Do We Compare Greek-Neutral Hedging Strategies?

- Delta-only hedging is widely used but insufficient in practice
- Market dynamics are driven by price movements, volatility changes, and time decay
- Hedging multiple risks introduces higher cost and implementation complexity

Research Question

Which Greek-neutral hedging strategy provides the best trade-off between hedging effectiveness and implementation cost across different volatility regimes?

Experimental Design

Setup: We keep the following settings fixed across all strategies



Underlying asset:
AMZN (non-dividend
paying stock)



Market:
Synthetic option
market



Rebalancing:
Daily



Execution engine:
Rolling-base
backtesting
framework



Portfolio setup:
Rolling ATM short
call position



**Shared option
panel:**
All strategies operate
on the same
persistent option
universe



Consistency:
Identical
assumptions across
all strategies

Ensures fair
comparison

Project Objective

Compare five hedging strategies under one common rolling-base backtest

Evaluate hedge quality, turnover, and funding burden under different volatility regimes

Identify the best practical trade-off between residual risk reduction and implementation cost

Methodology

Option Pricing and Core Greeks, under the European call option

Based on Black-Scholes Model

$$C = SN(d_1) - Ke^{-rT}N(d_2)$$

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \quad d_2 = d_1 - \sigma\sqrt{T}$$

$$\Delta = N(d_1) \quad \Gamma = \frac{\phi(d_1)}{S\sigma\sqrt{T}} \quad Vega = S\phi(d_1)\sqrt{T}$$

- Delta: Sensitivity to price
- Gamma: Sensitivity to delta
- Vega: Sensitivity to volatility

These sensitivities are computed for every live contract in the persistent option panel and drive the daily hedge solver.

Strategy Universe



Delta-neutral

Offsets first-order exposure to changes in the underlying price

Delta-Gamma-neutral

Reduces both first-order price risk and the sensitivity of delta to large price movements

Delta-Gamma-Vega

Controls gamma and volatility using multiple hedges, improving precision at higher cost

Delta-Vega-neutral

Reduces exposure to changes in implied volatility

Delta-Theta-neutral

Reduces exposure to time decay over the holding period

Methodology

Baseline Hedging Strategies

- Delta-neutral

$$n_S = -\Delta_P$$

- Delta-Gamma-neutral

$$q_H = -\frac{\Gamma_P}{\Gamma_H}$$

$$n_S = -(\Delta_P + q_H \Delta_H)$$

- Delta-vega-neutral

$$q_H = -\frac{V_P}{V_H}$$

$$n_S = -(\Delta_P + q_H \Delta_H)$$

- Delta-Theta-neutral

$$q_H = -\frac{\Theta_P}{\Theta_H}$$

$$n_S = -(\Delta_P + q_H \Delta_H)$$

Advanced Hedging Strategy

- Delta-Gamma-Vega

$$\begin{bmatrix} 1 & \Delta_a & \Delta_b \\ 0 & \Gamma_a & \Gamma_b \\ 0 & V_a & V_b \end{bmatrix} \begin{bmatrix} n_S \\ q_a \\ q_b \end{bmatrix} = - \begin{bmatrix} \Delta_P \\ \Gamma_P \\ V_P \end{bmatrix}$$

Baseline strategies use one hedge option and stock; DGV extends this with two hedge options and a constrained search.



Advanced Delta-Gamma- Vega (DGV) Hedging

The DGV framework starts with a short ATM option position sized against 100,000 underlying shares and rebalances daily.

Core Idea

Portfolio's exposure

$$(\Delta_P, \Gamma_P, V_P)$$

**Scaled Implementation
Matrix**

$$A = \begin{bmatrix} m\Gamma_a & m\Gamma_b \\ mV_a & mV_b \end{bmatrix}$$

**Neutralizing Remaining
Delta**

$$q_S = -(\Delta_P + m\Delta_a q_a + m\Delta_b q_b)$$

The Matrix System

$$\begin{bmatrix} \Gamma_a & \Gamma_b \\ V_a & V_b \end{bmatrix} \begin{bmatrix} q_a \\ q_b \end{bmatrix} = - \begin{bmatrix} \Gamma_P \\ V_P \end{bmatrix}$$

This is the key equation for simultaneously hedging Gamma and Vega using two options (a & b)

**Solve for Hedge Option
Quantities**

$$\begin{bmatrix} q_a \\ q_b \end{bmatrix} = -A^{-1} \begin{bmatrix} \Gamma_P \\ V_P \end{bmatrix}$$

And the Final Target

$$\Delta \approx 0, \quad \Gamma \approx 0, \quad V \approx 0$$

Why DGV method is more technical

Theoretical continuity meets the reality of discrete trading increments and operational constraints.

The Two-Layer Framework from four dimensions

1. Candidate Selection

Hedge options are chosen from

- a shorter-dated bucket
- a longer-dated bucket
- sometimes including currently held contracts

2. Condition Check

- the 2x2 gamma-vega matrix is tested using its condition number

3. Discrete Trade Search

- Search near the theoretical hedge and choose the trade that best minimizes the objective

The Multi-Objective Optimisation

The final trade selection is not a simple "zero-out" exercise, but a strategic trade-off framed as a Loss Function:

$$\mathcal{L} = \underbrace{w_{\Delta} \left(\frac{\Delta_{\text{res}}}{s_{\Delta}} \right)^2 + w_{\Gamma} \left(\frac{\Gamma_{\text{res}}}{s_{\Gamma}} \right)^2 + w_V \left(\frac{V_{\text{res}}}{s_V} \right)^2}_{\text{Residual Risk Tracking}} + \underbrace{\sum \lambda_k \cdot \text{Penalty}_k}_{\text{Friction \& Constraints}}$$

Key Optimization Factors:

- Pen_q (Size Deviation)
- Pen_t (Turnover)
- Pen_i (Inventory)

Evaluation Metrics



Hedge Effectiveness

$|\Delta|$ Absolute post-hedge
Delta

$|\Gamma|$ Absolute post-hedge
Gamma

$|\mathbf{V}|$ Absolute post-hedge
Vega

Implementation Cost

- Cumulative cash outflow (after interest)
- Turnover

Strategy Efficiency

- Residual risk vs turnover trade-off
- Hedge coverage across regimes

Market Regime and Execution Logic



Low volatility regime

- Calm Market
- Small price movements
- $VIX < 15$



Medium / normal regime

- Normal regime
- $VIX: 15-25$



High volatility regime

- Large moves / Jumps
- Event-driven periods
- $VIX > 25$



Volatility Regime Classification:

Low / Medium / High volatility



Regime Evaluation Day:

Day when market conditions are assessed and the current volatility regime is assigned



Hedge Eligible Day:

Day when the portfolio becomes eligible for hedging under the strategy rules



Hedge Executed Day:

Day when the hedge trade is actually implemented

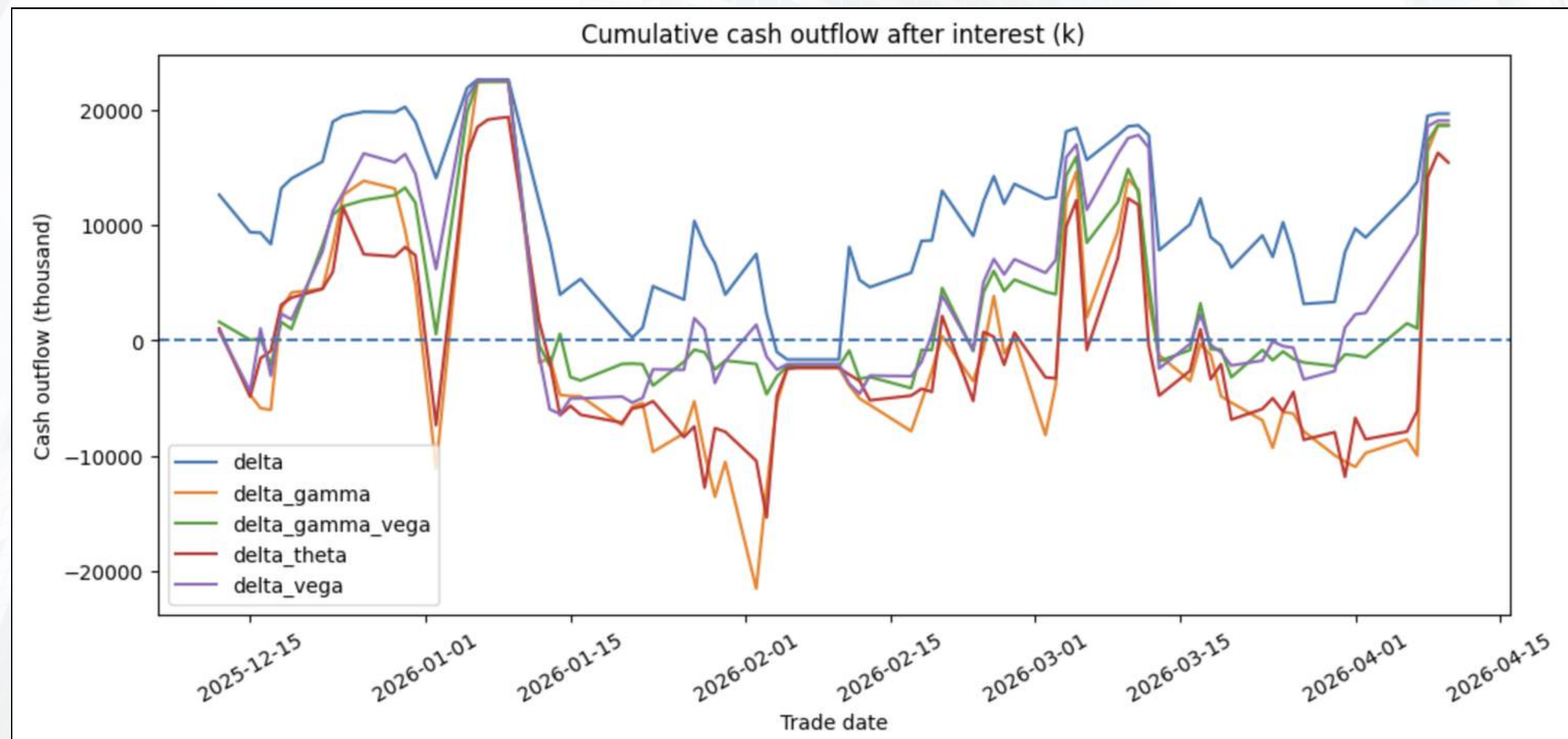
Why it matters:

- Regime labels are assigned within one continuous sample period
- Hedging is implemented once from end to end, rather than separately by regime
- This makes performance comparisons across market conditions more meaningful

Cumulative Cash Outflow (After Interest)

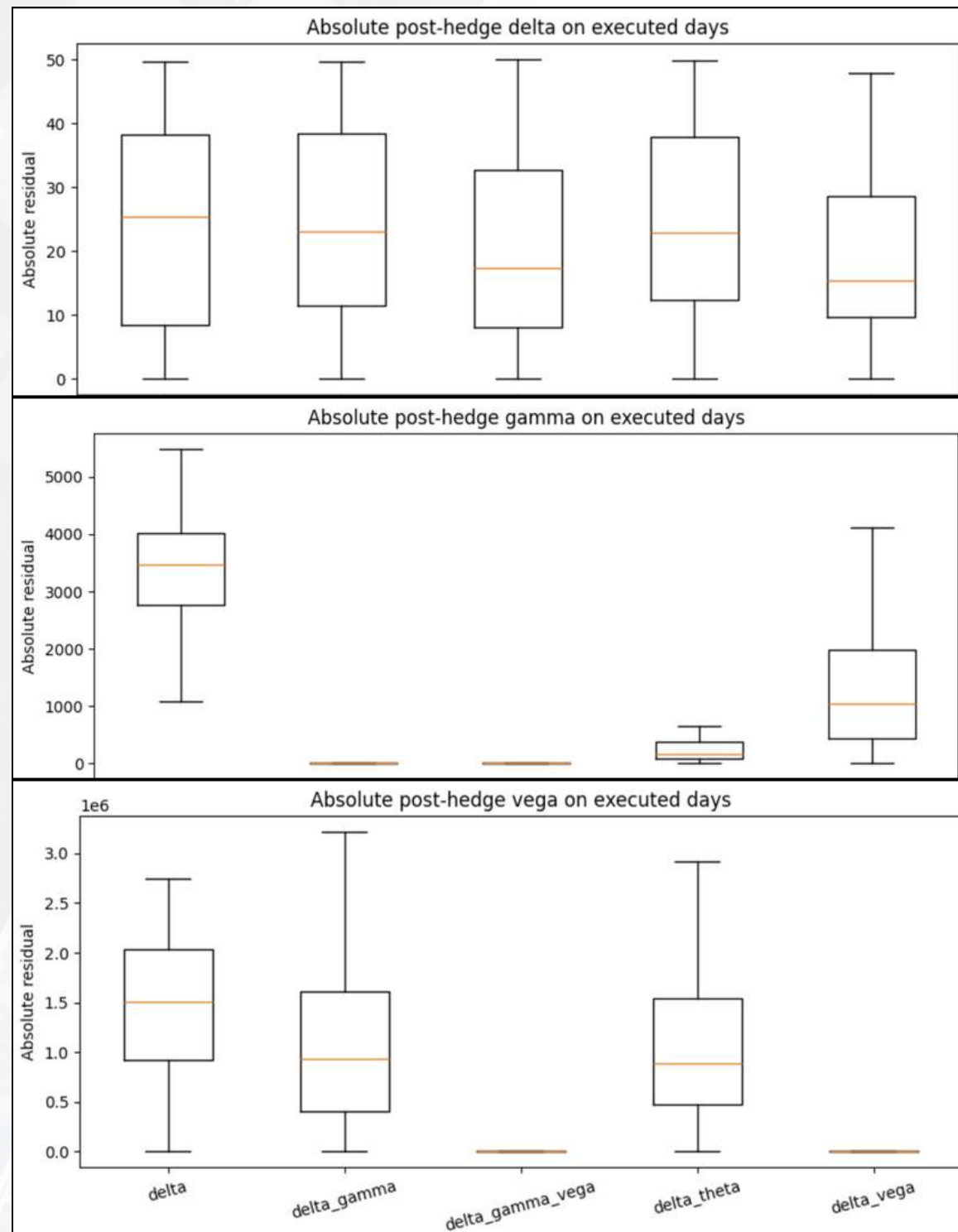
Delta-only hedging incurs the highest cost due to frequent rebalancing, while Delta-Gamma achieves the lowest and most stable implementation cost.

More complex strategies (Vega, DGV) improve hedge precision but introduce higher funding and trading costs.



Post-Hedge Risk Exposure

Hedge effectiveness improves with the number of Greeks neutralized



Delta (Δ):

Delta-Gamma and DGV reduce residual delta most effectively, while Delta-only and Theta leave higher first-order risk.

Gamma (Γ):

Delta-Gamma and DGV nearly eliminate gamma exposure, whereas Delta-only and Vega retain significant curvature risk.

Vega (V):

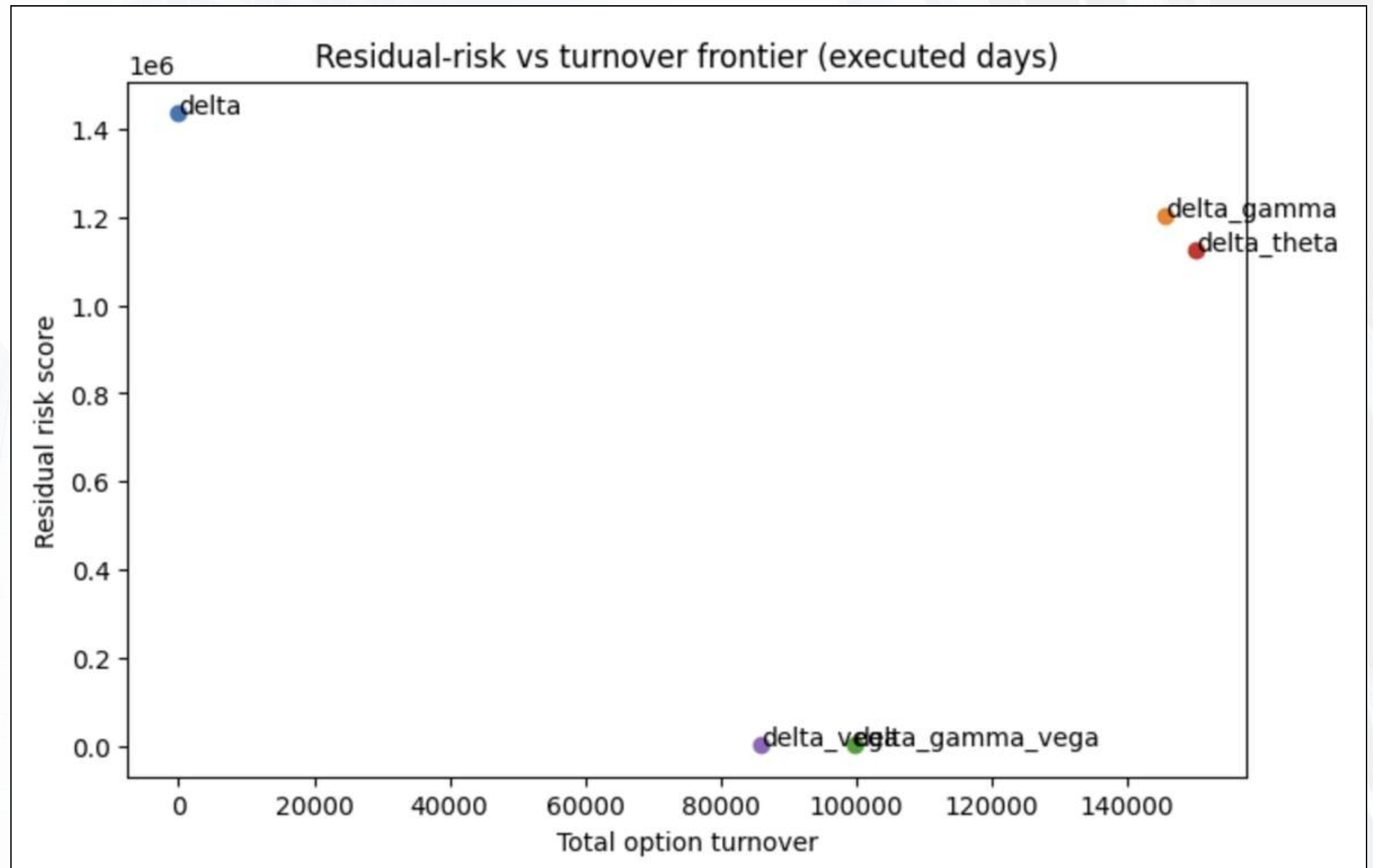
Vega and DGV effectively neutralize volatility exposure, while Delta and Gamma-based strategies leave residual vega risk.

Risk vs Turnover Trade-off

Delta-Vega and Delta-Gamma-Vega perform best overall, with Delta-Vega offering the stronger trade-off and DGV delivering the lowest residual risk.

Delta-Gamma and Delta-Theta are inefficient, with high turnover and high residual risk.

Delta-only has zero option turnover, but has the poorest hedge quality.



Ranking Strategies Across Regimes

=== HIGH VOLATILITY REGIME ===

	strategy	Post Δ	Post Γ	Post V	Turnover	Cash Outflow	Coverage %	HQ Rank	Cost Rank	Coverage Rank
0	delta_vega	19.91	1,578.24	31.96	16,662	9,255,838	100.0	1	2	1
1	delta_gamma_vega	30.18	1.17	1,550.94	20,820	1,055,006	100.0	2	1	1
2	delta	24.00	3,141.91	1,417,378.11	0	13,774,135	100.0	3	3	1
3	delta_theta	26.67	110.10	1,638,489.58	32,464	6,083,392	100.0	3	4	1
4	delta_gamma	28.83	0.05	1,785,305.68	30,379	9,984,762	100.0	3	5	1

=== LOW VOLATILITY REGIME ===

	strategy	Post Δ	Post Γ	Post V	Turnover	Cash Outflow	Coverage %	HQ Rank	Cost Rank	Coverage Rank
0	delta_gamma_vega	20.16	1.94	641.51	11,628	22,468,375	91.7	1	1	2
1	delta_vega	19.23	1,489.49	101.94	10,173	22,575,995	91.7	1	2	2
2	delta_gamma	25.35	0.10	1,188,424.55	19,489	22,478,272	91.7	2	3	2
3	delta_theta	29.07	608.25	1,246,365.03	21,344	19,378,042	100.0	3	2	1
4	delta	30.88	3,315.62	1,018,743.51	0	22,631,947	91.7	4	2	2

=== MEDIUM VOLATILITY REGIME ===

	strategy	Post Δ	Post Γ	Post V	Turnover	Cash Outflow	Coverage %	HQ Rank	Cost Rank	Coverage Rank
0	delta_gamma_vega	17.36	2.31	1,340.43	67,156	18,664,588	90.9	1	1	3
1	delta_vega	19.11	1,276.74	58.51	59,096	19,067,889	90.9	2	2	3
2	delta_gamma	22.48	0.08	1,046,210.35	95,714	18,665,048	92.7	3	3	2
3	delta_theta	22.94	231.69	962,837.71	96,281	15,437,062	96.4	4	2	1
4	delta	23.03	3,330.93	1,530,425.23	0	19,677,694	89.1	5	2	4

Comparison w.r.t.:

- **Hedge Quality (HQ):** Measured by Greeks' closeness to zero
- **Cost Effectiveness:** Measured with Options Turnover and Cash Outflow
- **Coverage:** hedge execution %

Comparing Strategies Across Regimes

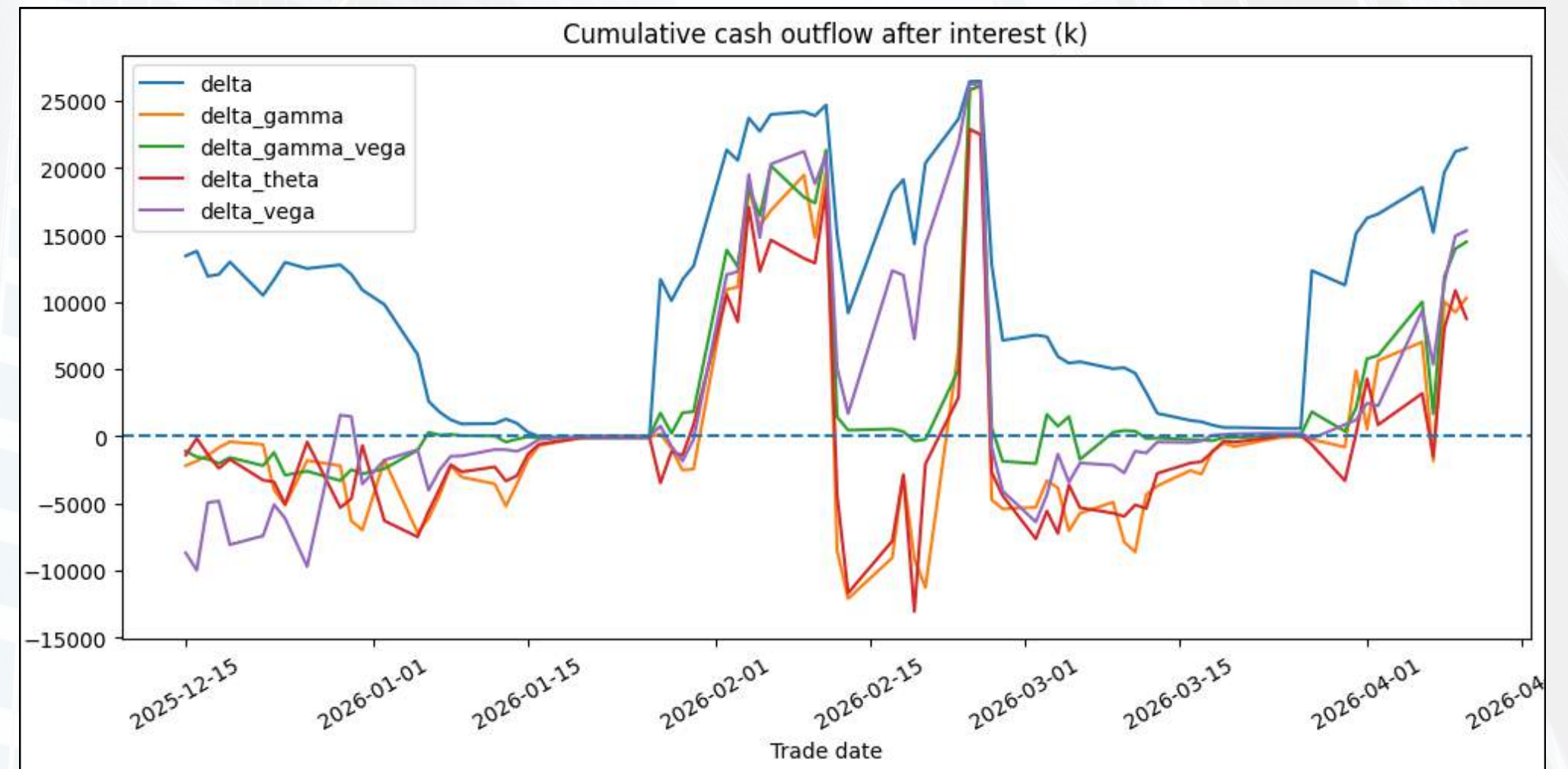
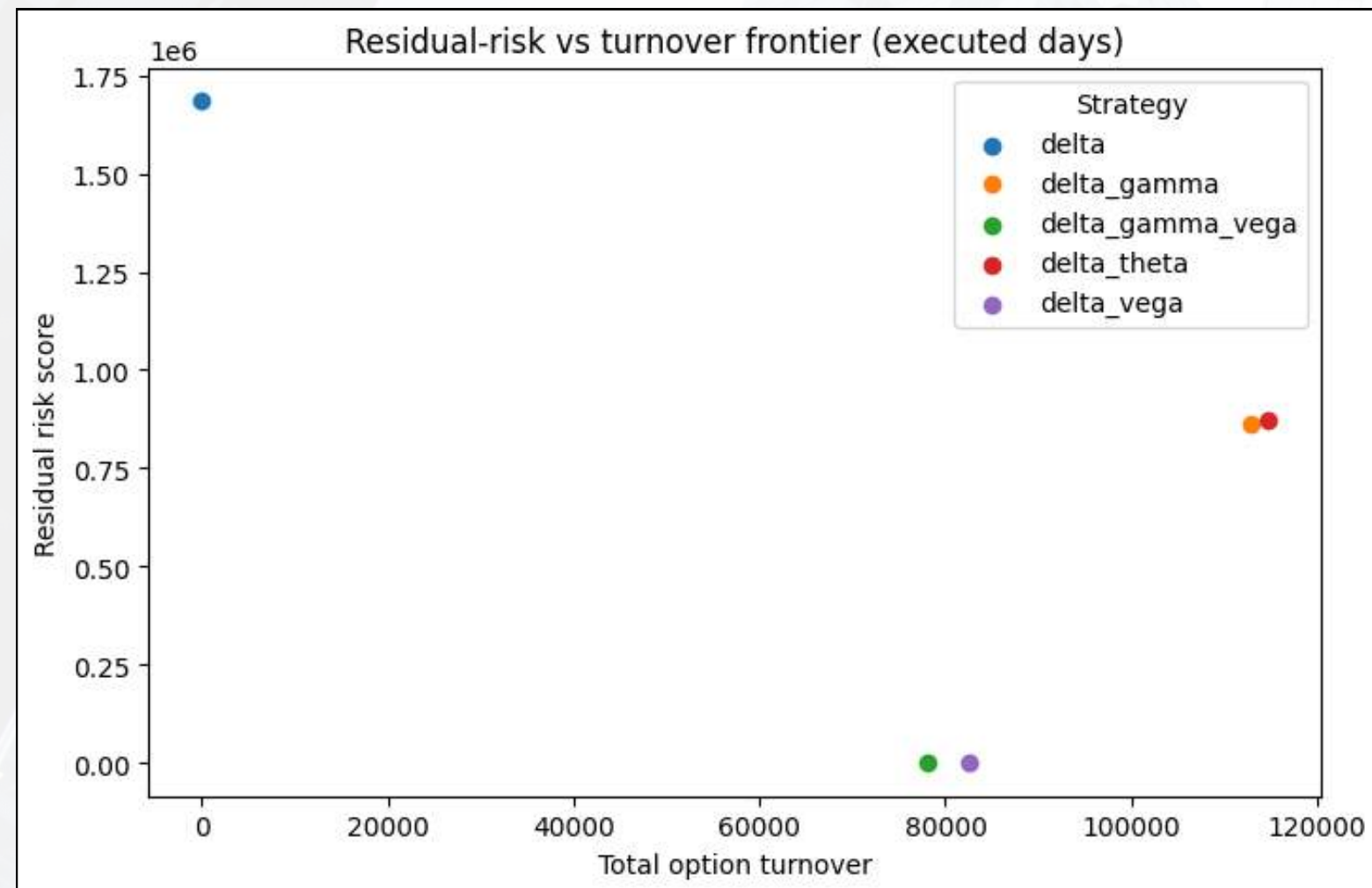
Volatility Regime	Best Hedge Quality	Best Cost Quality	Best Coverage	Best Overall Trade-off	Least Effective
High	Delta-Vega	Delta-Gamma-Vega	All	Delta-Vega	Delta-Gamma
Low	Delta-Gamma-Vega	Delta-Gamma-Vega	Delta-Theta	Delta-Gamma-Vega	Delta
Medium	Delta-Gamma-Vega	Delta-Gamma-Vega	Delta-Theta	Delta-Gamma-Vega	Delta

- Delta-Gamma-Vega is the strongest overall in low and medium volatility regimes, leading in hedge quality, cost efficiency, and overall trade-off.
- In high volatility, Delta-Vega performs best overall, suggesting that volatility-focused hedging becomes more important under stressed market conditions.
- Delta is consistently the weakest strategy, while Delta-Theta stands out mainly for hedge coverage rather than overall hedge effectiveness.

Effective hedging is not about eliminating all risks, but about efficiently targeting the dominant risks while controlling costs and execution complexity.

Testing with AAPL stock case

- We extended the study to AAPL, a dividend-paying stock, to test the robustness of the strategy comparison.
- The pricing pipeline was updated to include a continuous dividend yield in option pricing and Greeks, with no change to the hedging logic.
- The results remained broadly consistent and reaffirmed the effectiveness of Delta-Gamma-Vega hedging across market conditions.



Comparing Across Regimes (AAPL case)

= HIGH VOLATILITY REGIME ===									
strategy	Post Δ	Post Γ	Post V	Turnover	Cash Outflow	Coverage %	HQ Rank	Cost Rank	Coverage Rank
delta_gamma_vega	26.07	2.07	1,795.69	10,199	1,671,106	92.9	1	1	1
delta_vega	22.06	338.79	123.75	10,387	5,370,018	92.9	1	3	1
delta_theta	21.80	17.57	644,211.39	13,869	1,502,232	92.9	2	2	1
delta_gamma	32.36	0.06	391,015.02	13,014	1,841,388	92.9	3	3	1
delta	30.34	1,275.63	1,366,868.69	0	15,185,668	78.6	4	2	2
= LOW VOLATILITY REGIME ===									
strategy	Post Δ	Post Γ	Post V	Turnover	Cash Outflow	Coverage %	HQ Rank	Cost Rank	Coverage Rank
delta_vega	17.61	756.17	111.16	18,853	1,451,577	100.0	1	2	1
delta_gamma_vega	20.72	2.20	2,118.61	16,734	55,431	100.0	2	1	1
delta_theta	20.01	149.30	463,480.49	21,692	2,640,277	100.0	3	3	1
delta_gamma	23.00	0.06	614,370.98	22,517	3,032,975	100.0	4	4	1
delta	23.12	3,109.07	2,675,450.69	0	925,399	100.0	5	1	1
= MEDIUM VOLATILITY REGIME ===									
strategy	Post Δ	Post Γ	Post V	Turnover	Cash Outflow	Coverage %	HQ Rank	Cost Rank	Coverage Rank
delta_vega	21.44	849.42	80.48	53,337	15,312,999	92.6	1	3	2
delta_gamma_vega	23.45	1.89	1,692.84	51,224	14,485,696	94.4	2	1	1
delta_gamma	22.92	0.08	1,040,266.63	77,286	10,315,665	94.4	2	2	1
delta_theta	23.89	145.74	1,025,188.47	79,096	8,766,781	94.4	3	2	1
delta	20.61	2,059.16	1,513,095.11	0	21,470,898	92.6	3	2	2

Volatility Regime	Best Hedge Quality	Best Cost Quality	Best Coverage	Best Overall Trade-off	Least Effective
High	Delta-Gamma-Vega	Delta-Gamma-Vega	All except Delta	Delta-Gamma-Vega	Delta
Low	Delta-Vega	Delta-Gamma-Vega	All	Delta-Gamma-Vega	Delta-Gamma
Medium	Delta-Vega	Delta-Gamma-Vega	Delta-Gamma-Vega	Delta-Gamma-Vega	Delta

Verdict:

Delta-Gamma-Vega consistently delivers the best overall trade-off across regimes, while Delta-Vega performs strongly in hedge quality under normal and high volatility, and Delta remains the least effective strategy.

Limitations and Future works

Limitations

- Based on a synthetic options market; may not fully capture real-world microstructure, liquidity, or bid-ask effects.
- Analysis uses a single underlying (AMZN), limiting generalizability across assets and sectors.
- Market frictions are only partially modeled (turnover, funding); realistic factors like slippage, transaction costs, execution delays, and intraday constraints are not fully captured.

Future Works

- Extend to multiple underlyings and longer time horizons to test robustness of strategy rankings.
- Validate results using real-world historical options data.
- Incorporate richer execution frictions (bid-ask spreads, slippage, liquidity constraints, intraday effects) for more realistic implementation.

Conclusion

The best hedging strategy depends on the volatility regime and the balance between hedge quality, cost, and coverage.

Delta-Gamma-Vega is the strongest overall strategy in low and medium volatility regimes.

Delta-Vega performs best in high volatility, where volatility risk becomes more dominant.

Delta-only hedging is consistently the least effective across regimes.

Effective hedging should be judged by both risk reduction and practical implementation cost

Thank You.

Group 10

